

Abdullah Al Bashit

CURRICULUM VITAE

(U.S. Permanent Resident, authorized to work in the U.S.)

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Research Interests

I am broadly interested in developing statistical signal processing and machine learning methods to integrate diverse medical data, uncover the biology of complex diseases, and translate those insights into quantitative measures of disease risk and treatment response.

Education

Northeastern University, Boston, MA, USA

Jul 2019–Dec 2024

Ph.D. in Electrical Engineering

Thesis: Structural Characterization of Pathological Inclusions in Human Brain Tissue with Alzheimer's Disease.

Advisor: Prof. Lee Makowski

Texas State University, San Marcos, TX, USA

Aug 2017–Jun 2019

M.Sc. in Engineering (Electrical Engineering)

Thesis: A Comprehensive Solar Powered Remote Monitoring and Identification of Houston Toad Call Automatic Recognizing Device System Design.

Advisors: Prof. Damian Valles and Prof. Michael Forstner

Rajshahi University of Engineering & Technology (RUET), Bangladesh

Feb 2007–Jul 2011

B.Sc. in Electrical & Electronic Engineering

Thesis: Study of wavelength dependence of refractive index and material dispersion of various materials.

Advisors: Prof. S.M. Abdur Razzak and Prof. Md. Shahidul Islam

Research Experience

Computational Imaging Collaborator (*Neurovascular Segmentation*)

Apr 2026 – Present

Northeastern University, Dept. of Bioengineering, Boston, MA

- Created an end-to-end training and inference pipeline to segment blood-vessel structures in noisy 3D fluorescence microscopy images, handling weak contrast, uneven focus, broken structures, and limited annotated data.
- Fine-tuned an Attention U-Net with a pretrained ResNet34 encoder, using attention-gated skip connections to infer latent vessel structure, guided by engineered image features and augmentation.
- Trained and validated the model on an HPC cluster with 5-fold cross-validation, checkpointing, W&B tracking, attention-map review, and TP/FP/FN failure analysis, achieving approximately 91% Dice on held-out test data.

Postdoctoral Research Associate (*X-ray Scattering Signal Modeling*)

Jan 2025 – Present

Northeastern University, Dept. of Bioengineering, Boston, MA

- Developed a correlation-aware feature reduction method to extract weak biological signatures from noisy X-ray scattering data, reducing 211 candidate features to 11 disease-discriminative markers.
- Derived Bayes-risk and approximation-error bounds that determine pruning limits for highly correlated data while preserving discriminative signal.
- Trained a compact, interpretable neural network with 174 parameters to estimate pathological protein aggregation, achieving 84.3% test F1-score on held-out data.

Ancillary Co-Principal Investigator (*Multi-Omics Biomarkers of Lung Disease*)

Nov 2025 – Present

Brigham and Women's Hospital / Harvard Medical School, Boston, MA

- Lead transcriptomic analysis of COPD Gene participants ($n \approx 3,000$) to identify molecular signatures associated with mechanically affected lung (MAL) using participant stratification, differential gene expression, and pathway enrichment analysis.

- Develop proteomics mediation models to identify plasma proteins that mediate the relationship between mechanically affected lung (MAL) and 5-year emphysema progression.

Graduate Research Assistant (*Molecular Mapping of Alzheimer's Pathology*)

Oct 2020 – Dec 2024

Northeastern University, Dept. of Electrical & Computer Engineering, Boston, MA

- Designed an inverse-problem framework that generates Alzheimer's pathology maps from raw X-ray microdiffraction data, partnering with wet-lab teams at MGH for histology and biophysicists at Brookhaven National Laboratory for synchrotron measurements across 15 human brain samples.
- Formulated a pipeline that chains background subtraction, t-SNE, Gaussian Mixture Models, and a neural network to generate probabilistic spatial maps of pathological protein aggregates from noisy X-ray scattering measurements.
- Identified a consistent molecular signature of Alzheimer's cross- β fibrils: a ~ 4.7 Å β -strand spacing with an approximately -1° left-handed twist that persisted across disease stages.
- Authored 5 peer-reviewed publications and delivered 5 conference presentations, including 2 invited talks recognized with student lecturer awards.

Visiting Graduate Researcher (*Cryo-ET Automation*)

Sep 2022 – Dec 2022

Harvard Medical School, Dept. of Biological Chemistry & Molecular Pharmacology, Boston, MA

- Designed a Python-based GUI to automate cryo-electron tomography data acquisition, including microscope setup, motion correction, drift correction, and tilt-series alignment.
- Reduced setup time by 40% and increased throughput for structural biology workflows at Boston-area hospitals including Boston Children's Hospital, Dana-Farber, and MGH.

Graduate Research Assistant (*Wildlife Signal Detection and Sensing Device*)

Jan 2018 – Jun 2019

Texas State University, Dept. of Electrical Engineering, San Marcos, TX

- Prototyped and deployed a solar-powered Raspberry Pi acoustic monitor with onboard inference and cellular alerting for endangered Houston toad detection, scaling the system to 50 field units for conservation use in Bastrop County, Texas.
- Implemented power spectral density analysis, tuned bandpass filtering, MFCC extraction, and an LSTM classifier to distinguish Houston toad mating calls from other animal sounds and field noise, achieving 98% true-positive and 92.6% test accuracy on field audio.

Honors & Awards

• Student Travel Grant Award , ACA, Denver, Colorado.	2024
• Margaret C. Etter Student Lecturer Award , ACA, Baltimore, Maryland (2023) and Virtual (2021).	2023, 2021
• Ph.D. Network Travel Grant , Northeastern University.	2022–2024
• Ingram School of Engineering "Director's List" Award , Texas State University.	2018–2019
• Graduate College Scholarship , Texas State University.	2018
• Graduate Thesis Research Support Fellowship , Texas State University.	2018
• Bangladesh-Sweden Trust Fund Scholarship , Government of Bangladesh.	2018
• Ingram Graduate Scholarship , Texas State University.	2017
• 1st Place, IEEE Ethics Competition , SPAC, IEEE Region 10, Bangladesh.	2010
• Education Board Scholarship , Government of Bangladesh.	2007–2011

Technical Skills

Programming: C/C++, Python (scikit-learn, PyTorch), R, MATLAB, Bash.

Systems & Infrastructure: Linux/UNIX, SLURM, AWS, git.

Embedded/SoC: AVR, PIC, NXP, Arduino, Raspberry Pi.

Imaging & Structural Tools: X-ray scattering, UV microscopy, Fiji/ImageJ, QuPath.

Molecular Visualization: YASARA, Chimera, PyMOL.

Circuit Design/Simulation: PSPICE, Proteus.

Teaching

Guest Lecturer, Bioengineering, Northeastern University, Boston, MA. Apr 2026

BIOE 5410: Molecular Bioengineering (Host: **Prof. Lee Makowski**) – Delivered a guest lecture on machine learning for drug discovery to an audience of 52 undergraduate, M.S., and Ph.D. students.

Teaching Assistant, Northeastern University, Boston, MA. Sep 2020–Dec 2020

EECE 2323: Fundamentals of Digital Design and Computer Organization – Led recitation lectures and designed/conducted laboratory experiments for a class of approximately 40 undergraduate students.

Graduate Instructional Assistant, Texas State University, San Marcos, TX. Aug 2017–Dec 2017

MFGE 4392/4394: Microelectronics Manufacturing I/II – Supervised cleanroom laboratories on full CMOS-compatible semiconductor device fabrication (wafer preparation through final test) for a class of approximately 20 undergraduate students.

Lecturer, Prime University, Dhaka, Bangladesh. Feb 2012–Mar 2014

Delivered classroom lectures, designed and graded exams, supervised labs, and held office hours for classes of approximately 20–80 undergraduate students.

- EEE 157: Electrical Circuit II (Fall 2012)
- EEE 210: Electronic Circuit Simulation Lab (Spring 2012)
- EEE 235: Energy Conversion I (Summer 2012)
- EEE 325: Energy Conversion II (Summer 2012, Spring 2013)
- EEE 326: Energy Conversion II Lab (Spring 2013)
- EEE 407: Microprocessor Systems and Interfacing (Spring 2012)
- EEE 425: Power System I (Summer 2012, Spring 2013)
- EEE 431: Power System II (Fall 2012)
- EEE 473: Power System Protection (Fall 2012)

Mentorship

Mentored 7 students across B.S., M.S., and Ph.D. degree programs in Bioengineering, Bioinformatics, and Electrical Engineering.

- Tom Pu, PhD candidate in Department of Bioengineering (NEU, 2027) 2026
- Helena Zheng, BS in Bioengineering and Biochemistry (NEU, 2026) 2026
- Ganesan Murugan, MS Department of Bioinformatics (NEU, 2026) 2026
- Durga Gomathi Arumuganainar, MS Department of Bioinformatics (NEU, 2026) 2025–2026
- Wendao Li, BS+MS Department of Bioengineering (NEU, 2026) 2024
- Devyn Stringfellow, BS in Bioengineering and Biochemistry (NEU, 2026) 2023
- Fairuz Samiha Saeed, MS Department of Electrical Engineering (TXSTATE, 2021) 2020–2021

Professional Memberships & Service

- **Workshop Instructor**, Technical Skills Series, Department of Bioengineering, Northeastern University, Boston, MA. 2026
Designed curriculum, presentation materials, and hands-on training sessions for M.S. and Ph.D. students in Biology, Bioengineering, Biochemistry, and related fields.
 - Python for Data Analysis & Automation (15 participants)
 - Applied Machine Learning for Biomedical Data (11 participants)
- **Graduate Student Member**, American Crystallographic Association (ACA). 2021, 2023, 2024

- **Graduate Student Member**, Biophysical Society (BPS). 2022, 2024
- **Graduate Student Member**, IEEE, Texas State University. 2018–2019
- **Student Coordinator**, IEEE RUET Student Branch. 2009–2011
- **Trainer**, Workshop on PIC and AVR microcontrollers at RUET, Bangladesh. 2009

Industry Experience

Assistant Engineer, Bangladesh Power Development Board (BPDB), Bangladesh Apr 2014–Aug 2017
Designed and tested electrical power distribution transformers and line accessories.

Peer-Reviewed Journal Publications

- P. Nepal, **A. A. Bashit**, [4 authors], and L. Makowski, “Scanning X-ray microdiffraction studies of protein accumulation and tissue loss in thin sections of human brain tissue in Alzheimer’s disease,” *submitted*, May 2026, arXiv preprint pending.
- **A. A. Bashit**, P. Nepal, and L. Makowski, “A Multicollinearity-Aware Signal-Processing Framework for Cross- β Identification via X-ray Scattering of Alzheimer’s Tissue,” *under review*, Nov. 2025. [Preprint: [arXiv](#)]
- P. Nepal, **A. A. Bashit**, and L. Makowski, “Characterization of sub-micrometre-sized voids in fixed human brain tissue using scanning X-ray microdiffraction,” *Journal of Applied Crystallography (IUCr)*, Oct. 2024.
- P. Nepal, **A. A. Bashit**, L. Yang, and L. Makowski, “Small-angle X-ray microdiffraction from fibrils embedded in tissue thin sections,” *Journal of Applied Crystallography (IUCr)*, Dec. 2022.
- J. Martinez-Lorenzo, J. Hudack, Y. Jing, M. Shaham, Z. Liang, **A. A. Bashit**, [5 authors], and A. Fox, “Preliminary Experimental Results of Context-Aware Teams of Multiple Autonomous Agents Operating under Constrained Communications,” *Robotics*, Sep. 2022.
- **A. A. Bashit**, P. Nepal, T. Connors, D. H. Oakley, B. T. Hyman, L. Yang, and L. Makowski, “Mapping the Spatial Distribution of Fibrillar Polymorphs in Human Brain Tissue,” *Frontiers in Neuroscience*, May 2022.
- F. S. Saeed, **A. A. Bashit**, V. Viswanathan, and D. Valles, “An Initial Machine Learning-Based Victim’s Scream Detection Analysis for Burning Sites,” *Applied Sciences*, Sep. 2021.
- M. R. Islam, G. K. Beng, and **A. A. Bashit**, “Design and Fabrication of Segment Display Architecture for Displaying Bengali and English Numerals,” *Australian Journal of Basic and Applied Sciences (AJBAS)*, Nov. 2014.
- A. A. Mansur, **A. A. Bashit**, M. R. Islam, “Harmonic Analysis of Front-End Current of Three-Phase Single-Switch Boost Converter,” *International Journal of Applied Information Systems (IJ AIS)*, Mar. 2013.

Peer-Reviewed Conference Publications

- **A. A. Bashit** and D. Valles, “MFCC-based Houston Toad Call Detection using LSTM,” *Proceedings of the 2019 IEEE International Symposium on Measurement and Control in Robotics (ISMCR)*, Houston, Texas, USA, Sep. 2019.
- **A. A. Bashit** and D. Valles, “A Solar Powered Raspberry Pi Houston Toad Call Detection System Using Neural Network Model,” *Proceedings of the 2018 International Conference on Computational Science and Computational Intelligence (CSCI)*, Las Vegas, Nevada, USA, Dec. 2018.
- **A. A. Bashit** and D. Valles, “A Mel-Filterbank and MFCC-based Neural Network Approach to Train the Houston Toad Call Detection System Design,” *Proceedings of the 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, British Columbia, Canada, Nov. 2018.
- **A. A. Bashit** and D. Valles, “An Embedded Approach for Controlling Automatic Water Pump and Monitoring Real-Time Remote Data on Desktop, Android, and Web-based Application,” *Proceedings of the 16th International Conference on Embedded Systems, Cyber-physical Systems, and Applications (ESCS)*, Las Vegas, Nevada, USA, Aug. 2018.

Conference Presentations

- **A. A. Bashit**, P. Nepal, and L. Makowski, “Integrating semi-supervised learning and transformer neural networks for lesion classification in Alzheimer’s disease pathology,” *American Crystallographic Association (ACA) 74th Annual Meeting*, Denver, Colorado, USA, Jul. 2024. [Oral Presentation]
- **A. A. Bashit**, P. Nepal, and L. Makowski, “Probing the microstructure of pathological protein deposits in Alzheimer’s disease,” *Biophysical Society (BPS) 68th Annual Meeting*, Philadelphia, Pennsylvania, USA, Feb. 2024. [Oral Presentation]
- **A. A. Bashit**, P. Nepal, L. Makowski, and L. Yang “Mapping of pathological inclusions in human brain tissue with Alzheimer’s disease,” *American Crystallographic Association (ACA) 73rd Annual Meeting*, Baltimore, Maryland, USA, Jul. 2023. [Oral Presentation]
- **A. A. Bashit**, P. Nepal, and L. Makowski, “Unsupervised learning approach for mapping of pathological lesions in X-ray scanning microdiffraction studies of Alzheimer’s disease,” *Biophysical Society (BPS) 66th Annual Meeting*, San Francisco, California, USA, Feb. 2022. [Oral Presentation]
- **A. A. Bashit**, P. Nepal, and L. Makowski, “Classification of tissue variations in X-ray scanning microdiffraction from thin sections of human brain,” *American Crystallographic Association (ACA) 71st Annual Meeting*, Virtual, Aug. 2021. [Oral Presentation]

Invited Talks

- “Discovering the Hidden Secrets: Understanding Pathological Inclusions in Alzheimer’s Brain Tissue, Systems Pharmacology,” Brigham and Women’s Hospital (BWH), Sep. 2025.
- “Classification of tissue in X-ray scanning microdiffraction from thin sections of human brain,” Brookhaven National Laboratory (BNL), Mar. 2023.

Posters

- P. Nepal, **A. A. Bashit**, T. R. Stewart, B. T. Hyman, and L. Makowski, “Mapping of the molecular architecture of pathological protein deposits in Alzheimer’s disease,” *Biophysical Society (BPS) 69th Annual Meeting*, Los Angeles, California, USA, Feb. 2025.
- P. Nepal, **A. A. Bashit**, and L. Makowski, “Small-angle X-ray microdiffraction from fixed human brain tissue exhibits power-law behavior that provides insights into the structural organization of neuropathological lesions,” *Biophysical Society (BPS) 68th Annual Meeting*, Philadelphia, Pennsylvania, USA, Feb. 2024.

In the News

- **Northeastern BioE GSC Mentorship Hub**, “Featured as Workshop Instructor for the Technical Skills Series: Applied Machine Learning and Python Data Automation” (2026).
- **Newswise**, “Understanding Alzheimer’s Progression with Improvements to Imaging, Image Processing, Machine Learning” (2021).
- **SmugMug**, “Drone demonstrations at Air Force Pitch Day” (2019).